Reconstructing compact-binary mass-posterior orientation from the mass-ratio marginal

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The component-mass posterior of a compact-binary gravitational-wave event is elongated along a contour of near-constant chirp mass, a shape usually treated as a nuisance degeneracy. We show that its *orientation* is quantitatively reconstructible from a single one-dimensional marginal of the same posterior. Evaluating the constant-chirp-mass curve over an event's own mass-ratio marginal predicts the principal-axis position angle to a median 1.26° on GWTC-4.0/O4a and 1.22° on GWTC-5.0/O4b for elongated posteriors (axis ratio ≥ 3), with no coefficient calibrated on either catalog and with the prediction fixed on GWTC-3 beforehand. The local tangent approximation used by rapid parameter-estimation tools achieves 4.20–6.67° on the same events. This is not a curve merely fitting a curved distribution: substituting any other event's marginal degrades the error by 3.2–11.3×, and the achieved error falls below the minimum of 300 catalog-stratified permutations in all three catalogs. Inputs from one waveform family recover another family's independently sampled axis, so the agreement does not arise from shared Monte Carlo noise. That the constant-chirp-mass exponents specifically carry the information is shown directly: neighbouring exponents evaluated on the same marginal are 2.9–4.7 times worse, with a best-fitting exponent slightly above the general-relativistic value. That small offset is not a deviation from general relativity but the imprint of the next post-Newtonian order: a parameter-free Fisher calculation predicts an effective exponent of 0.630, matching the data. The residual $\sim 1^\circ$ is 17.2 times the Monte Carlo resolution of the released samples and is therefore a property of the model. We present this as a measurement of posterior geometry and a systematics diagnostic, not as a test of general relativity.

I. INTRODUCTION

Parameter estimation for a compact-binary coalescence yields a posterior probability distribution over the source parameters. In the component-mass plane this distribution is characteristically elongated: the chirp mass

$$\mathcal{M}_c = \frac{(m_1 m_2)^{3/5}}{(m_1 + m_2)^{1/5}} \tag{1}$$

dominates the leading-order inspiral phase and is measured far more precisely than the mass ratio, so the posterior extends along a contour of near-constant $\mathcal{M}_c$ [1, 2, 4]. This shape is routinely described qualitatively and treated as a degeneracy to be marginalised over.

We ask whether the shape carries quantitative, predictable structure. Specifically: can the orientation of the (m_1, m_2) posterior — the position angle of its principal axis — be recovered event by event from low-dimensional summaries of the same posterior, using no coefficient calibrated on the data being predicted, and does such a reconstruction survive on later, disjoint catalogs?

It can, and the construction is elementary. Evaluate the constant-$\mathcal{M}_c$ curve over the event's own mass-ratio marginal, and take the principal axis of the resulting set of points. That axis predicts the posterior's own principal axis to about a degree on elongated posteriors, in two catalogs released after the prediction was fixed.

The reason it works is visible in Figure 3. Posterior samples are spread along an *arc*, not a line, so their principal axis is a chord of that arc and is rotated away from the tangent at the median point. Using the arc rather than the tangent supplies essentially all of the accuracy: the same comparison run with the tangent is several times worse. The organising idea is that the uncertainty geometry of gravitational-wave inference is largely *measurement optics* — a property of what the signal and instrument can resolve rather than of source astrophysics — and that this geometry is precise enough to be used as a diagnostic.

The ingredients are individually familiar. That inference problems possess a few stiff and many sloppy directions is the content of the sloppy-model framework [8]; that the principal axis of data spread along a curved arc differs from the local tangent is the distinction between linear principal component analysis and a *principal curve* [9]. A complementary strategy is to remove such degeneracies by reparameterisation rather than describe them: Makinen et al. [7] detect degenerate parameter combinations from the Fisher information and search for symbolic coordinate transformations that flatten it, which reduces the simulation cost of neural posterior estimation. We do the opposite, characterising the degeneracy in the standard mass coordinates in which posteriors are released and used. The published work closest to ours stops short in identifiable ways. Hannam et al. [4] state that "the component masses consistent with the signal lie roughly along a line of constant chirp mass" and label chirp mass on their confidence regions, but make no quantitative statement about orientation. Ohme et al. [5] apply prin-

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cipal component analysis to post-Newtonian phase coefficients rather than to (m_1, m_2) posterior samples, predict no per-event orientation, and perform no catalog validation. The `simple-pe` metric of Fairhurst et al. [6] assumes the match "varies quadratically with the difference in parameters" about a peak, yielding an error ellipse — the same local, linear approximation as our tangent baseline, though computed in $(\mathcal{M}, \eta)$ rather than (m_1, m_2) coordinates — and those authors note that they "observe curved contours rather than perfect ellipses". We are not aware of published work that reconstructs per-event posterior orientation from the chirp-mass curve over an event's own mass-ratio marginal, or that validates such a reconstruction out-of-sample across catalogs.

Section II defines the construction and the data. Section III reports the out-of-sample performance and the tests that establish it is not trivial. Section IV examines the origin of the residual, and Section V its robustness to coordinate choice and to the elongation threshold. Section VI discusses scope and limitations.

II. METHOD AND DATA

A. The predicted orientation

For an event with source-frame posterior samples $\{(m_1, m_2)\}$, let $\psi_{\rm meas}$ be the position angle of the principal axis of the sample covariance and let the axis ratio axr be the ratio of its principal standard deviations. The constant-$\mathcal{M}_c$ curve in the mass plane is

$$m_1(q) = \mathcal{M}_c\,(1+q)^{1/5} q^{-3/5}, \qquad m_2 = q\,m_1, \tag{2}$$

with $q = m_2/m_1 \in (0, 1]$ and $\mathcal{M}_c$ the event's median source-frame chirp mass. The predicted orientation $\psi_{\rm curve}$ is the principal axis of the point set obtained by evaluating Eq. (2) over the event's own mass-ratio marginal. The local linear baseline $\psi_{\rm tangent}$ is the tangent to the constant-$\mathcal{M}_c$ contour at the posterior median point. No coefficient is calibrated on the validation catalogs; every ingredient is supplied by the event itself.

B. The prediction depends only on the mass-ratio marginal

The construction has one input, not two. Changing $\mathcal{M}_c$ multiplies $m_1(q)$ and $m_2(q)$ by a common factor, which is a dilation about the origin; a dilation leaves the eigenvectors of the sample covariance unchanged. The predicted orientation is therefore *exactly* invariant to $\mathcal{M}_c$, and $\psi_{\rm curve}$ is a functional of the mass-ratio marginal alone.

The chirp mass remains necessary to place the curve in the mass plane, which the thickness and self-consistency analyses of Section IV and the geometry shown in Figure 3 require, but it contributes nothing to the orientation. One consequence is used in Section V A: repeating the analysis in detector-frame masses changes the *measured* axis only, never the prediction.

This is a statement about the internal geometry of a single posterior, not a prediction from information external to it: the mass-ratio marginal is drawn from the same posterior whose orientation is reconstructed, and no information from the (m_1, m_2) covariance is used. What is demonstrated is that a two-dimensional posterior's orientation is recoverable from one of its own one-dimensional marginals. Section III B establishes that the chirp-mass contour specifically, and the event's own marginal specifically, are what carry that recovery.

C. Data and event selection

We use public parameter-estimation samples from three releases: GWTC-2.1 [16] and GWTC-3 [17], which together form the derivation set; GWTC-4.0/O4a [11]; and GWTC-5.0/O4b [12]. The two later catalogs are mutually disjoint and disjoint from the derivation set, sharing no events. Samples are required to be finite in all mass parameters and to satisfy $0.02 < q \leq 1$.

The analysis is restricted to *elongated* posteriors, axr $\geq$ 3. This is a physical requirement rather than a tuned threshold: as a covariance approaches isotropy its principal axis ceases to be defined, and any orientation prediction must degrade toward a random baseline. Section V B shows that the score varies smoothly across the threshold and that nothing distinguishes the chosen value.

The relation was fixed on the derivation set and registered before any later-catalog file was opened; the scores in Table I are the resulting locked decision. Appendix D describes the computational pipeline and the provenance of every number reported here.

III. RESULTS

A. Out-of-sample performance

On elongated events the median mismatch $|\psi_{\rm curve} - \psi_{\rm meas}|$ is 1.26° on O4a and 1.22° on O4b. These are the locked values and are used in every headline statement. The figures recompute them independently from the cached extract of Appendix D: the recomputation agrees on O4a (1.26°), and on O4b gives 1.19° against the locked 1.22°, the two differing only in which waveform group is taken as preferred for a small number of events. We report both rather than reconcile them.

Table I collects the locked scores together with the tangent baseline and the replication statistics. Every cell is a locked, full-sample value except the derivation-catalog win fraction, for which no locked value exists; that one is computed from the cached extract and is marked in the caption. Figure 1 shows the per-event comparison behind those medians, drawing each elongated event as a line joining its tangent error to its curve error so that the full distribution is visible rather than a summary.

B. The chirp-mass exponents carry the information

A curve describing a curved distribution better than a straight line is unsurprising, and a construction with no free parameters cannot overfit. The substantive questions are whether the constant-$\mathcal{M}_c$ contour *specifically* carries the information, and whether the *event's own* marginal does. Both hold.

First, the exponents matter. Generalising the contour to $m_1(q;p) \propto q^{-p}(1+q)^{2p-1}$, with general relativity fixing $p = 3/5$, and evaluating each alternative on the *same* event's own mass-ratio marginal, displacing the exponent to $p = 0.50$ degrades the median orientation error to 3.4–4.1°, a factor of 2.9–4.7 worse than the general-relativistic value across the three catalogs. The error as a function of p has a well-defined minimum, at $p^\star = 0.62$–0.63: slightly *above* 3/5. That small displacement is real and physical, and Section IV D shows it is the imprint of the next post-Newtonian order rather than a defect of the construction.

The improvement over the local tangent is likewise not an artefact of an implausibly crude comparison. A chord joining the contour at the 5th and 95th percentiles of the marginal — a one-parameter stand-in for the arc — already improves substantially on the tangent, reaching 2.78° on O4a and 1.89° on O4b; but the full arc improves on the chord by roughly a further factor of two, so the curvature of the contour across the marginal carries information beyond its endpoints.

Second, the event's own marginal is required (Figure 2): substituting any other event's does not perform comparably.

Replacing the event's marginal with one pooled over the catalog, or with another event's, degrades the median error by 3.2–11.3×. The band shown in Figure 2 is that null distribution; it is not a confidence interval on the reconstruction. Under 300 catalog-stratified permutations of the mass-ratio marginals among events, the achieved error lies below the *minimum* of the null distribution in every catalog, giving a permutation p-value $< 1/300$ throughout. We report the full permutation distribution rather than a single shuffled assignment because one draw is not a null: in O4a the 300 draws span 3.19–10.35°, so any individual shuffle could have fallen anywhere in that range.

C. Transfer between waveform families

Because the mass-ratio marginal is taken from the posterior being reconstructed, a natural concern is circularity: the agreement might reflect shared Monte Carlo noise rather than shared geometry. We test this by crossing waveform families. Using family A's mass-ratio marginal to predict family B's independently sampled principal axis — never touching B's own geometry — gives 2.08° (A→B) and 2.78° (B→A) over 64 events elongated in both, against tangent baselines of 3.57° and 6.25° in the same directions. The reference scale is 1.99°, the median angle between the two families' own measured axes.

The reconstruction therefore transfers between separately sampled posteriors of the same event and is not an artifact of reusing one posterior's Monte Carlo fluctuations. The test has a definite limit: the two families analyse the same strain data with shared calibration and priors, so it excludes reuse of sampling noise but not modelling error common to both. We describe 1.99° as a reference scale rather than an error floor, since a reconstruction may in principle denoise a posterior functional and fall below the raw disagreement between families.

IV. ORIGIN OF THE RESIDUAL

A. The residual exceeds the Monte Carlo resolution

The $\sim 1°$ residual is not sampling noise. Bootstrapping the released samples — drawing one index set and recomputing *both* the measured axis and the prediction inputs from it, so that their correlated Monte Carlo errors cancel — gives a median resolution of 0.07° on the difference, against a median discrepancy of 1.07°. Because the bootstrap is computed per event, the appropriate summary is the median of the per-event ratios, 17.2; this is not the ratio of the two medians just quoted, which is a different statistic. The constant-$\mathcal{M}_c$ curve is an accurate but imperfect description of posterior orientation, and the gap is a property of the model rather than of the sample count.

This bootstrap measures the Monte Carlo resolution of the released sample set, and is distinct from the interval drawn in Figure 1, which is a bootstrap over the event sample. It shrinks as more samples are released and encodes nothing about detector noise, calibration, waveform systematics or population variation; it is not a coverage statement, and we do not report the fraction of events falling within one or two bootstrap widths as though it were.

The residual carries a signed component: the curve under-rotates relative to the measured axis by a median 0.75°. This is not uniform across catalogs, being significant in the derivation catalog ($p < 0.001$) and in O4a ($p = 0.019$) but not in O4b ($p = 0.110$), the newest and largest of the three. We therefore report it per catalog.

B. Failure of principal-curve self-consistency

Hastie & Stuetzle [9] define a principal curve by *self-consistency*: $f(t) = \mathbb{E}[X \mid t_f(X) = t]$, so that every point of the curve is the mean of the data projecting onto it.

	GWTC-3 (derivation)	GWTC-4.0/O4a	GWTC-5.0/O4b
median $\lvert\psi_{\mathrm{curve}} - \psi_{\mathrm{meas}}\rvert$ (elongated)	—	**1.26°**	**1.22°**
median $\lvert\psi_{\mathrm{curve}} - \psi_{\mathrm{meas}}\rvert$ (all)	—	7.53°	5.41°
median $\lvert\psi_{\mathrm{tangent}} - \psi_{\mathrm{meas}}\rvert$ (all)	7.49°	9.36°	8.23°
chirp-vs-total win fraction	78%†	86%	88%
Spearman($\lvert\Delta\psi_{\mathrm{tangent}}\rvert$, axr)	−0.42	−0.40	−0.69
N events, total	75	86	104
N events, elongated	—	19	32

TABLE I. Locked out-of-sample scores, computed on the full released posterior samples. The reconstruction lands within $\sim 1.2°$ of the measured orientation on elongated events in two later, disjoint catalogs. The tangent error *decreases* with elongation (final row), i.e. orientation becomes better defined for more elongated posteriors; normalised arc length behaves the same way, correlating negatively with the tangent error in all three catalogs ($\rho = -0.51, -0.57, -0.49$). †No locked win fraction exists for the derivation catalog, so that cell is computed from the cached extract; the other two are locked.

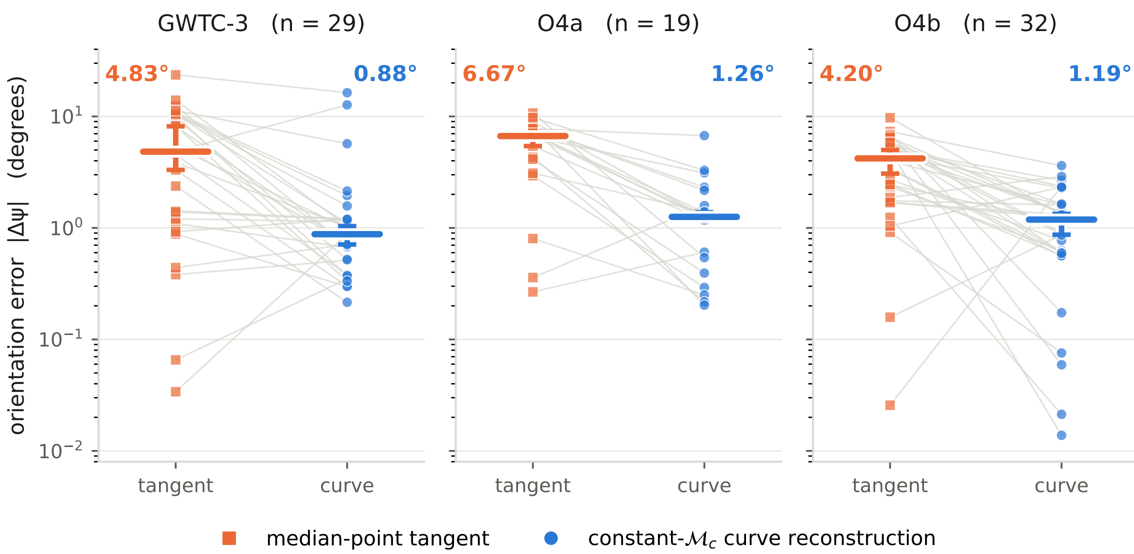

FIG. 1. Reconstruction error against the local linear approximation, on elongated events (axr ≥ 3) in three disjoint catalogs. Each thin line is one event, joining its median-point tangent error to its constant-$\mathcal{M}_c$ curve error; heavy bars are catalog medians with a bootstrap interval over events. The curve improves on the tangent in every catalog: GWTC-3 ($n = 29$): $4.83° \to 0.88°$, O4a ($n = 19$): $6.67° \to 1.26°$, O4b ($n = 32$): $4.20° \to 1.19°$. Note the logarithmic scale.

This is directly testable for the constant-$\mathcal{M}_c$ curve, and it fails. Projecting each posterior sample onto its nearest curve point and comparing that point with the mean of the samples landing there leaves a perpendicular offset of about 2.6% of the posterior's own scale: the constant-$\mathcal{M}_c$ curve is not a principal curve of the posterior.

The magnitude of that failure tracks the reconstruction residual. The per-event self-consistency violation correlates with the per-event orientation residual at Spearman $\rho = 0.68$ ($p = 7 \times 10^{-12}$) on the primary projection grid, with ρ between 0.68 and 0.69 across a sweep of grid resolutions, so the result does not depend on that choice. Both quantities are measured rather than fitted, so the correlation cannot be inflated by additional model freedom.

We describe this as tracking rather than explaining. Both quantities derive from the same posterior and the same curve, so the correlation localises the residual within the geometry without establishing a mechanism, and the corresponding correction does not survive out-of-sample. Replacing the curve by the conditional means — a single Hastie–Stuetzle iteration — reduces the residual in sample, but the improvement is grid-sensitive (median 0.59–1.09° across the sweep) and in the cross-family test it improves both directions while reaching $p < 0.05$ in only one ($1.20° \to 0.64°$, $p = 0.002$; $0.92° \to 0.57°$, $p = 0.107$). We therefore do not present it as an improved reconstruction.

C. Finite posterior thickness

A second contributor is supported out-of-sample. The curve model has zero thickness, whereas a real posterior

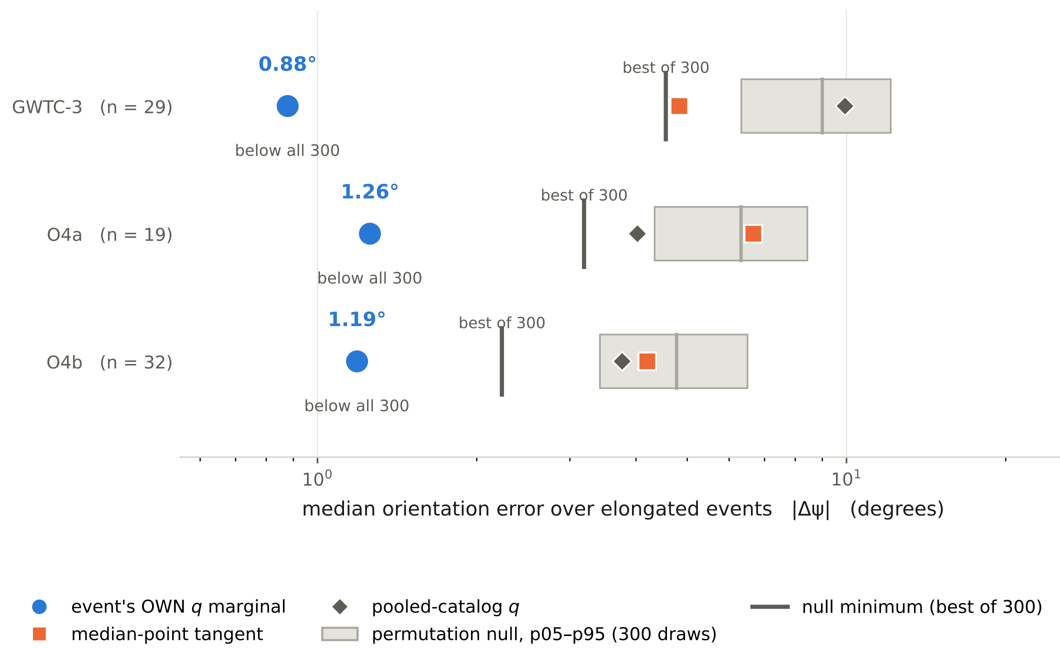

FIG. 2. The reconstruction requires the *event's own* mass-ratio marginal. Blue circles use the event's own q marginal, orange squares the median-point tangent, grey diamonds a q marginal pooled over the catalog. The grey band spans the 5th–95th percentile of a permutation null in which each event is given a different event's q marginal (300 catalog-stratified draws); the dark tick marks that null's minimum. Own-q falls below the null minimum in every catalog (GWTC-3: own-q 0.88° vs. null minimum 4.55°; O4a: own-q 1.26° vs. null minimum 3.19°; O4b: own-q 1.19° vs. null minimum 2.23°).

has finite width perpendicular to the curve — a median of 0.34 of its total spread. Learning that width profile on one waveform family and using it to predict another family's axis improves on the zero-thickness curve in both directions ($1.20° \to 0.91°$, $p = 3 \times 10^{-4}$; $0.92° \to 0.72°$, $p = 0.008$).

The *arc-varying* part of that profile is not established: a constant thickness performs as well as the measured profile in one direction, and simple linear or quadratic tapers match or exceed it in both. We therefore attribute no specific fraction of the residual to the measured taper.

D. A post-Newtonian origin for the exponent offset

The displacement of the best-fitting exponent slightly above 3/5 (Section III B) has a physical explanation that also bears on the residual. At leading (0PN) order the inspiral phase depends on the masses only through $\mathcal{M}_c$, so the degenerate direction of the posterior is exactly the constant-$\mathcal{M}_c$ contour. The next (1PN) term introduces a dependence on the symmetric mass ratio η, which rotates that direction slightly and shifts the effective exponent above 3/5.

We make this quantitative. Constructing the stationary-phase inspiral Fisher matrix in (m_1, m_2) over the inspiral band and taking its least-constrained direction — using no information from the measured posterior — reproduces the constant-$\mathcal{M}_c$ tangent exactly at leading order (effective exponent 0.600), and at 1PN predicts an effective exponent of 0.630 (interquartile range 0.628–0.639). This parameter-free prediction matches both the empirical optimum above ($p^\star = 0.62$–0.63) and the exponent fitted independently in Appendix A. The offset is therefore the imprint of 1PN phasing.

The rotation is a *local* effect and accounts for only part of the residual: at the median point it moves the predicted orientation from 4.9° (leading order) to 4.5° (1PN), while the full arc reaches $\sim$1°. Most of the accuracy comes from integrating the curvature of the contour across the marginal, not from the local direction; the remaining residual is shared between that arc curvature and the finite posterior thickness of Section IV C.

V. ROBUSTNESS

A. Frames and coordinates

The measured quantity is a Euclidean principal-axis angle in fixed coordinates and is not reparameterisation invariant. We state that dependence rather than assume it away. Repeating the primary score in detector-frame masses gives 0.32° against 1.07° in the source frame, paired over the same 80 events ($p = 6 \times 10^{-9}$); in $(\log m_1, \log m_2)$ it gives 0.77°. In $(\mathcal{M}_c, q)$ coordinates the construction is degenerate by design, since a constant-

$\mathcal{M}_c$ curve is a vertical line and the reconstruction carries no per-event content.

The detector-frame improvement is largely mediated by elongation rather than indicating a better-suited model. Removing the redshift uncertainty that source-frame masses inherit makes posteriors markedly more elongated (median axis ratio 9.8 against 4.6), and at matched axis ratio the two frames no longer separate cleanly: in the 3–5 band the detector frame is in fact worse (5.51° against 1.37°). The source frame remains the locked primary score and the detector-frame comparison is post hoc.

B. The elongation threshold

The restriction to elongated posteriors is physical rather than tuned. As the covariance approaches isotropy the principal axis ceases to be defined and the score degrades smoothly toward the random baseline: the median error is 14.6° for axr $\in [1, 1.5)$, then 7.3°, 5.8°, 1.4° and 0.6° for successive bands, approaching but never reaching the 45° expected for a random angle modulo 180°. Nothing distinguishes the chosen threshold of 3: the error falls monotonically and smoothly across thresholds from 1.5 to 10. Figure 3 illustrates the geometry for three events selected by stated rules, including the worst case in the sample. The reconstruction does not always win: the local tangent achieves the smaller error for 15% of elongated events, rising to 31% above axr = 8, where both approaches are already accurate and the arc correction is small. Any application relying on the reconstruction should treat it as a typical rather than a guaranteed improvement.

C. Dependence on source parameters

If the geometry is measurement optics, the residual should not depend on source astrophysics. We test this directly. Ranking the 32 elongated O4b events by their curve residual and correlating against nine pre-declared axes — precession χ_p, aligned spin $\chi_{\rm eff}$, mass ratio, redshift, signal-to-noise ratio, chirp mass, total mass, secondary mass, and waveform-family disagreement — with Benjamini–Hochberg control and a partial correlation removing the axis-ratio confound, no physical axis survives. The only surviving correlate is waveform-model disagreement ($\rho = -0.64$, $q = 7 \times 10^{-4}$), a measurement systematic, with a sign indicating that the curve tracks the waveform *ensemble* consensus more closely than any single waveform tracks another.

This is a null result on 32 events and has correspondingly limited power. It bounds the size of any physical dependence; it does not exclude one.

VI. DISCUSSION

The orientation of a compact-binary mass posterior is reconstructible from a single one-dimensional marginal of that posterior to about a degree on elongated events, with no coefficient calibrated on the validation catalogs, and the reconstruction holds out-of-sample on two later, disjoint catalogs. It is not the trivial observation that a curve outperforms a line: substituting another event's marginal degrades the result several-fold, the achieved error lies below the minimum of 300 permutations, and the reconstruction transfers between separately sampled waveform-family posteriors of the same event. The residual is a systematic of the model, 17.2 times the Monte Carlo resolution of the released samples, to which finite posterior thickness is an out-of-sample contributor.

The tangent baseline quantifies what is being corrected. The principal axis of a Gaussian is the leading eigenvector of its covariance, which for a likelihood expanded quadratically about its peak is the local tangent to the degeneracy; the tangent error is therefore the error incurred by treating the posterior as Gaussian. On elongated events that error is 4.92° against 1.07° once the arc is accounted for — a factor of 4.6. For applications that use posterior orientation, the arc correction is essentially free: it requires the mass-ratio marginal, which parameter estimation already produces.

a. Scope. Three boundaries are worth stating together, since each has a natural stronger reading that the measurements do not support. First, the relation uses a marginal of the posterior it reconstructs, not information external to it: the mass-ratio marginal is drawn from that same posterior. Second, the reconstructed quantity is a Euclidean angle in fixed coordinates, and Section V A shows it changes with the choice of mass variables; we therefore report the convention rather than claim invariance, and do not appeal to Čencov's uniqueness theorem for the Fisher–Rao metric, which would assert the opposite of what we measure. Third, we use the stiff/sloppy vocabulary [8] for the concept only. The stronger "hyperribbon" picture, in which the covariance eigenvalue spectrum spans many orders of magnitude, does not follow from these data: in the two-dimensional mass plane the eigenvalue ratio has a median of 3.3, with 28% of events exceeding one order of magnitude and 3% exceeding two. Sloppiness in that sense is a property of the full parameter space, which a two-dimensional marginal cannot establish.

The posteriors analysed here were generated with general-relativistic waveform models, so structure found in their geometry probes the internal consistency of those inferences rather than the theory that produced them, and no bound on modified inspiral phasing follows. Appendix A develops this quantitatively for a one-parameter generalisation of the chirp-mass exponents, where a naive reading gives an apparent 3.1σ departure from the general-relativistic value that two pre-committed checks reduce to 1.5σ.

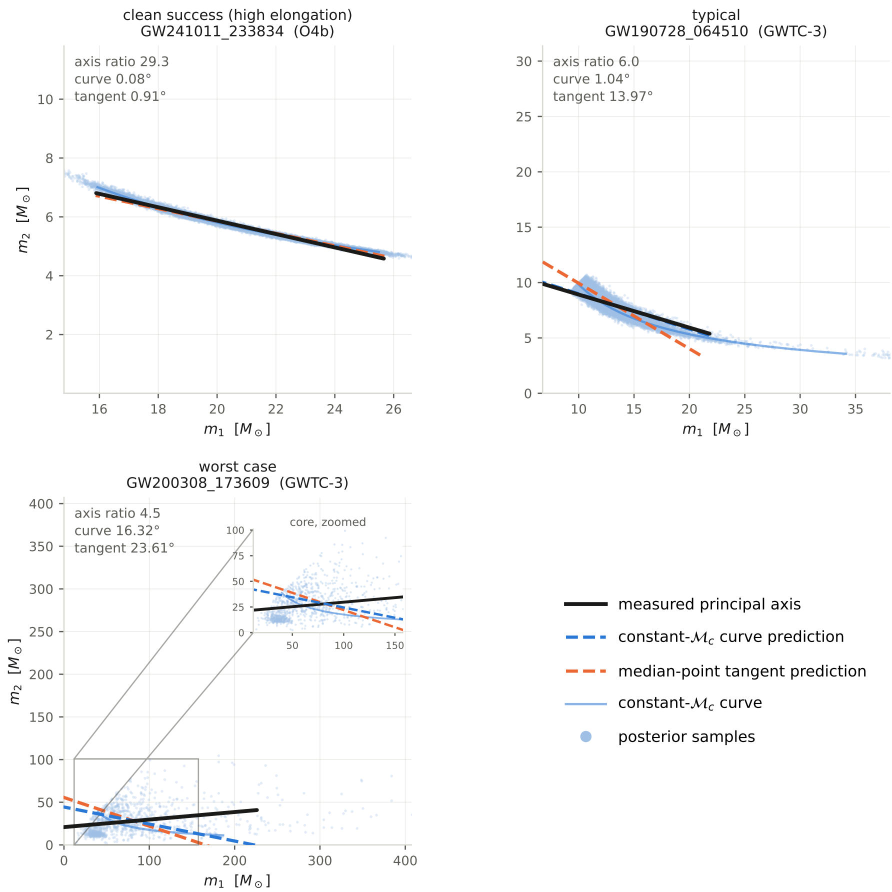

FIG. 3. Posterior geometry for three events chosen by fixed rules — *clean success (high elongation)* GW241011_233834 (O4b, axr 29.3, curve 0.08°, tangent 0.91°); *typical* GW190728_064510 (GWTC-3, axr 6.0, curve 1.04°, tangent 13.97°); *worst case* GW200308_173609 (GWTC-3, axr 4.5, curve 16.32°, tangent 23.61°). Each panel shows the posterior samples, the constant-$\mathcal{M}_c$ curve, the measured principal axis and the two predicted orientations, on equal-aspect axes. The worst case is included by rule rather than chosen, and carries a zoom inset on its dense core; its failure is genuine, the error remaining at 16.2–17.5° when the sample is trimmed to its central 90%. Across all elongated events the tangent beats the curve for 15% of events, rising to 31% above axr = 8.

b. Limitations. We record the following, roughly in order of how much they constrain use of the result.

1. Waveform-model choice contributes a per-event orientation scatter of several degrees and is the dominant per-event systematic.
2. The relation is validated on elongated posteriors. Round posteriors have no well-defined orientation and are excluded by construction, and Baird et al. [3] note that constant chirp mass characterises the mass degeneracy less cleanly at higher masses, bounding where the construction should be expected to hold.
3. O4a and O4b are disjoint event catalogs, not independent experiments: they share detectors, calibration, waveform families, priors and analysis conventions.
4. The two out-of-sample scores are not equal evidence. The O4b registration is timestamped in the public record before the data were opened; the O4a registration entered the public repository in the same commit as its results and is attested only by a private history.
5. The residual is tracked by the curve's failure of self-consistency, but the corresponding one-step correction does not survive out-of-sample, and the arc-varying form of the thickness contribution is not established.
6. A first-principles derivation of the curve's width and curvature from the Fisher information remains open.

VII. CONCLUSIONS

The orientation of the component-mass posterior of a compact binary is not merely qualitatively aligned with a constant-chirp-mass contour but quantitatively reconstructible from the event's mass-ratio marginal, to a median 1.26° and 1.22° on two catalogs released after the prediction was fixed. The reconstruction requires the event's own marginal, transfers between waveform families, and leaves a residual that is a property of the model rather than of finite sampling.

Two consequences follow. First, the local Gaussian approximation underlying rapid parameter-estimation tools misstates posterior orientation by a factor of 4.6 more than the arc-corrected construction, which suggests a cheap correction for applications that rely on the orientation. Second, because the geometry is this reproducible, departures from it are measurable, and we find no dependence on any of nine pre-declared source parameters — the only surviving correlate is waveform-model disagreement. A first-principles derivation of the curve's thickness and curvature from the Fisher information, and a strain-level rather than model-level test of the underlying information geometry, remain open.

Appendix A: A geometric exponent diagnostic

The exponents in Eq. (2) are fixed by the leading-order quadrupole formula. Generalising to a one-parameter family,

$$\mathcal{M}_p = (m_1 m_2)^p (m_1 + m_2)^{1-2p}, \tag{A1}$$

with $p = 3/5 = 0.600$ in general relativity. Fitting p to the posterior orientations isolates the exponent from the mass scale, since orientation is scale-free ($m_1(q;p) \propto q^{-p}(1+q)^{2p-1}$).

On the O4b elongated events we find $p = 0.616 \pm 0.011$; combining O4a and O4b gives $p = 0.628 \pm 0.009$, formally 3.1σ from $p = 3/5$. This is not a deviation from general relativity, and it is not a statistical fluctuation to be explained away: it is the imprint of the next post-Newtonian order, predicted from first principles in Section IV D. The 1PN η-dependent phasing shifts the effective degeneracy exponent to 0.630, in agreement with the fitted value here and with the empirical optimum of Section III B.

Two further properties confirm that the offset is a feature of the leading-order geometric model rather than new physics, and both were pre-committed. The two disjoint catalogs disagree with each other by 0.031, comparable to the offset from the general-relativistic value (0.028), so the catalog-to-catalog spread is a systematic that the bootstrap statistical error (0.009) does not capture; including it gives 1.5σ. And the per-event exponent depends on elongation (Spearman $= -0.41$), with the most-elongated half yielding $p = 0.606$, consistent with 3/5 as the finite-width effect that produces the offset diminishes. No bound on modified inspiral phasing follows from any of this, and we draw none: the posteriors were generated with general-relativistic waveforms.

Appendix B: Model-level information anatomy behind the measurement-optics view

The measurement-optics reading of Section I makes a further prediction that can be checked against the waveform model directly: the same Fisher structure that fixes the posterior's orientation also fixes *when*, in frequency, each parameter's information accumulates. Computing the cumulative Fisher information for chirp mass (0PN), mass ratio (1PN) and spin (1.5PN) across 190 O4a and O4b events, the ordering $\mathcal{M}_c < \eta < \chi$ holds universally: chirp mass is determined earliest (~ 35 Hz), spin latest, and the frequency separation between them is a monotonic function of total mass (Spearman -1.00), widest for light, long-inspiral systems and compressed for heavy systems that merge promptly in band.

The ordering is a property of the waveform model, computed from it rather than measured from strain, and the Spearman value reflects that determinism rather than a statistical result. It therefore illustrates the framework rather than testing it, and the results of Sections III–V do not depend on it. A data-driven version, re-analysing loud events in frequency slices and comparing empirical with predicted information accumulation, would turn this into a measurement and is left to future work.

Appendix C: Why signal-to-noise and orientation informativeness differ: GW250114

The quantity reconstructed in this paper requires a well-defined principal axis, which is a property of posterior *shape* rather than of loudness. The two can come apart sharply, and the loudest event detected to date is the clearest example. The ringdown results quoted below are other groups' work and are included for orientation.

GW250114 is a near-equal-mass binary black hole detected at network signal-to-noise ~ 77, with remnant mass $M_f^{\rm src} \approx 62.8\,M_\odot$ and spin $\chi_f \approx 0.68$ — essentially a much louder counterpart of GW150914. Its remnant implies Kerr quasi-normal-mode frequencies of 248.9 Hz ($\tau = 4.10$ ms) for the fundamental and 243.4 Hz ($\tau = 1.35$ ms) for the first overtone. A published ringdown analysis using an orthonormal-mode basis reports the first overtone at 99.9% significance — up from 82.5% in nonorthogonal analyses of the same event — and finds "no significant deviation" from the Kerr prediction [10]. We quote the orthonormal figure with that context because the significance is a property of the analysis basis, and we describe the Kerr result as the null it is reported to be rather than as a confirmation of the no-hair theorem.

Being near-equal-mass, its posterior is nearly round, and it therefore falls outside the elongated sample used throughout this paper despite being the loudest event available. Loudness buys precision on the parameters; it does not buy the anisotropy that an orientation measurement needs. Selecting events for a geometric probe by signal-to-noise would therefore be the wrong criterion. We make no ringdown claim of our own for this event. An independent ringdown analysis attempted during this work was withdrawn when its posterior was found to be prior-dominated, the credible interval reproducing the prior rather than measuring the data, and we therefore rely on the published analyses cited above.

Appendix D: Reproducibility

All analyses read a single cached extract of the released posterior samples, so that every downstream result shares one deterministic input and can be regenerated without re-reading the parameter-estimation files. The extract performs no subsampling: it stores every sample passing the finiteness and mass-ratio cuts, 23.1 million samples across 972 waveform-group rows, with 972 of those 972 rows held in full. A cache-backed number is therefore a full-sample number and carries no resampling noise.

As a consistency check, an independent pass that rereads the released files directly, bypassing the cache, returns 1.26° for O4a and 1.19° for O4b, against 1.26° and 1.19° from the cache. For the derivation catalog the two differ by 0.08°, because they resolve one event's preferred waveform group differently; this is a selection convention rather than a sampling effect.

Every number in this paper — in the abstract, the text, Table I and every figure caption — is emitted from a committed machine-readable artifact and inserted into the manuscript source as a generated macro rather than typed, so that text, table and figure cannot drift from the analysis or from one another. Figure captions are generated from the same sidecar files that produce the figures.

DATA AVAILABILITY

This work is a reanalysis of public parameter-estimation data. It uses no proprietary or embargoed products and generates no new observational data. Three releases are used, obtained from the Gravitational Wave Open Science Center (GWOSC) [13] and Zenodo: GWTC-2.1 [16] and GWTC-3 [17] posterior samples; the GWTC-4.0/O4a release [11], Zenodo record 16053484; and the GWTC-5.0/O4b release [12], Zenodo records 20276106 and 20348006. GWOSC data are distributed under a Creative Commons Attribution licence and are used here under those terms. Analysis code, registered predictions, per-event results and the figure-generation pipeline accompany this work; record numbers and download procedures are given in the accompanying data-availability documentation.

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This is independent work carried out entirely on public data. The author is not a member of the LIGO–Virgo–KAGRA Collaboration; the Collaboration has not reviewed this analysis and bears no responsibility for its conclusions.

a. Use of AI assistants. Large language models were used in this work and are disclosed here in the form journal policy now requires. *Tools:* Anthropic's Claude (Opus 4.8, via Claude Code) and OpenAI's Codex. *How they assisted:* Claude implemented the analysis code and the figure and number-generation pipelines and drafted the manuscript; Codex performed independent adversarial review of both the analysis and the text over repeated rounds. *How they were directed:* by the author, through task-level instructions specifying the measurement to be made, the preregistered decision rules to be respected, and the artifacts against which each claim was to be checked; neither model had discretion over what was claimed or which results were reported. *How output was verified:* every number in this paper is regenerated from a committed machine-readable artifact rather than transcribed (Appendix D), a contract test suite fails if manuscript and artifacts disagree, and the author reviewed each result before inclusion. The author defined the research questions, registered every preregistered decision before the corresponding data were examined, and is solely responsible for the content of this paper. The models are tools, not authors.

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